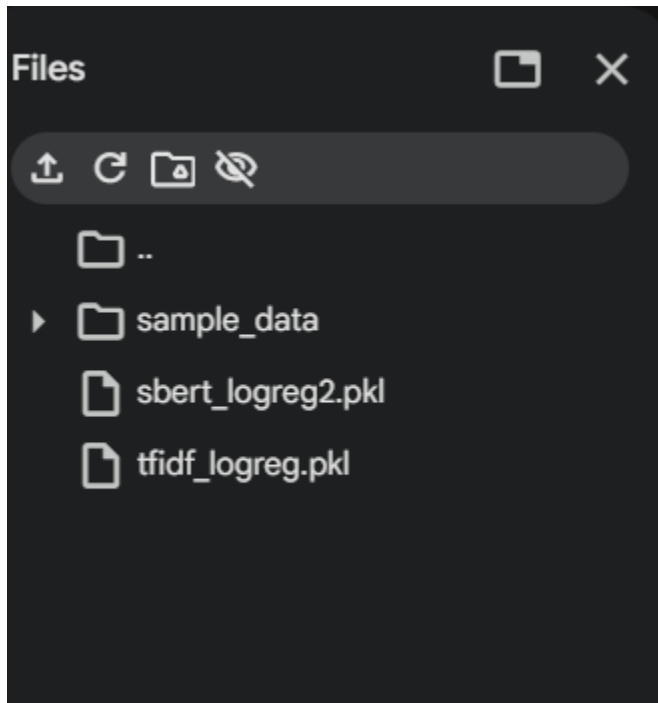


How to run our simple Gradio application:

1. Open our code in Google Colab
2. Install dependencies (gradio, scikit-learn, sentence-transformers, nltk). It is preferable that this version of Gradio is installed (more stable), `pip install gradio==4.31.5 -q datasets sentence-transformers scikit-learn joblib nltk`
3. Upload all model .pkl files into the environment.



4. Run the final cellblock in our code after installing all the dependencies
5. These results should pop up once u run the program

```
Loaded sbert_logreg2.pkl type: <class 'dict'>
sbert_clf: <class 'sklearn.linear_model._logistic.LogisticRegression'>  sbert_scaler: <class 'NoneType'>
Loaded tfidf_logreg.pkl type: <class 'dict'>
tfidf_vec: <class 'sklearn.feature_extraction.text.TfidfVectorizer'>  tfidf_clf: <class 'sklearn.linear_model._logistic.LogisticRegression'>  pipeline
Launching Gradio...
Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://079fcad6b9db4bbf0b.gradio.live
```

6. Click on the public url to test our application
7. Test it however you like with any kind of tweet!

SBERT vs TF-IDF Sentiment (robust)

Enter text

I hate this movie, it sucks.

Analyze

SBERT result

Label (SBERT)

Negative 😞

Confidence (SBERT)

0.9976

Probabilities (SBERT)

[0.9976, 0.0024]

TF-IDF result

Label (TF-IDF)

Negative 😞

Confidence (TF-IDF)

0.7058

Probabilities (TF-IDF)

[0.7058, 0.2942]